Fundamentals

1. What is Data Analysis?

Answer:

Meaning:
Data Analysis is the process of collecting, cleaning, and studying data to find useful information and patterns.

- Purpose:
  It helps in making better decisions, like understanding customer behavior or improving business performance.
- Steps involved:
  Data collection → Data cleaning → Data processing → Interpretation → Conclusion.
- Example:
  Analyzing sales data to find which product sells the most.

2. How to Plan a Data Science Project (All Steps)

Answer:

- 1. Problem Understanding:
  Clearly define what problem you want to solve (e.g., predicting customer churn).
- 2. Data Collection:
  Gather data from sources like CSV files, databases, APIs, etc.
- 3. Data Cleaning & Preprocessing:
  Handle missing values, remove duplicates, and convert data into usable format.
- 4. Data Analysis & Visualization:
  Use graphs and charts to understand patterns (Power BI, Python, etc.).
- 5. Model Building:
  Apply machine learning algorithms (like Logistic Regression, Decision Trees).
- 6. Model Evaluation:
  Check accuracy using metrics like precision, recall, or confusion matrix.
- 7. Deployment & Reporting:
  Present results using dashboards or reports and deploy the model if needed.

3. Machine Learning Problem Statement (Customer Churn)

Answer:

- Problem Definition:
  Predict whether a customer will stop using a service (churn) based on their purchase behavior.
- Input (Features):
  Customer data like purchase history, frequency, amount spent, subscription duration.
- Output (Target):
  Churn = Yes (1) or No (0).
- Goal:
  Help businesses retain customers by identifying those likely to leave.

4. What are Tensors?

(a) What are Tensors?

- Definition:
  Tensors are multi-dimensional arrays used to store data in machine learning and deep learning.
- Types of Tensors:
  - Scalar → Single number (0D)
  - Vector → 1D array
  - Matrix → 2D array
  - Higher dimensions → 3D, 4D, etc.
- Usage:
  Used in frameworks like TensorFlow and PyTorch to handle large datasets.
- Example:
  Image data is stored as a 3D tensor (height × width × color channels).

(b) In-depth Explanation with Simple NumPy Example

- NumPy Representation:
  Tensors can be created using NumPy arrays.
- Examples:

```
import numpy as np

# Scalar (0D tensor)
a = np.array(5)

# Vector (1D tensor)
b = np.array([1, 2, 3])

# Matrix (2D tensor)
c = np.array([[1, 2], [3, 4]])

# 3D Tensor
d = np.array([[[1, 2], [3, 4]], [[5, 6], [7, 8]]])
```

- Key Idea:
  As dimensions increase, tensors can store more complex data like images, videos, etc.
- Why Important:
  They allow efficient computation and are the backbone of deep learning models.